

1. 求下列规划问题的K-T点

$$\begin{cases} \min & f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2 \\ \text{s.t.} & g_1(x_1, x_2) = x_1^2 + x_2^2 - 5 \leq 0 \\ & g_2(x_1, x_2) = x_1 + 2x_2 - 4 \leq 0 \\ & g_3(x_1, x_2) = -x_1 \leq 0 \\ & g_4(x_1, x_2) = -x_2 \leq 0 \end{cases}$$

$$L(x, \mu) = (x_1 - 3)^2 + (x_2 - 2)^2 + \mu_1(x_1^2 + x_2^2 - 5) + \mu_2(x_1 + 2x_2 - 4) + \mu_3(-x_1) + \mu_4(-x_2)$$

KKT条件:

$$\frac{\partial L}{\partial x_1} = 2(x_1 - 3) + 2\mu_1 x_1 + \mu_2 - \mu_3 = 0$$

$$\frac{\partial L}{\partial x_2} = 2(x_2 - 2) + 2\mu_1 x_2 + 2\mu_2 - \mu_4 = 0$$

$$x_1^2 + x_2^2 - 5 \leq 0$$

$$x_1 + 2x_2 - 4 \leq 0$$

$$-x_1 \leq 0$$

$$-x_2 \leq 0$$

$$\mu_1, \mu_2, \mu_3, \mu_4 \geq 0$$

$$\mu_1(x_1^2 + x_2^2 - 5) = 0$$

$$\mu_2(x_1 + 2x_2 - 4) = 0$$

$$\mu_3(-x_1) = 0$$

$$\mu_4(-x_2) = 0$$

假设 $\mu_3 = 0, \mu_4 = 0$ 。则 g_1 和 g_2 为起作用约束:

$$x_1^2 + x_2^2 - 5 = 0$$

$$x_1 + 2x_2 - 4 = 0$$

由 $x_1 + 2x_2 - 4 = 0$ 得 $x_1 = 4 - 2x_2$ 。

代入 $x_1^2 + x_2^2 - 5 = 0$:

$$(4 - 2x_2)^2 + x_2^2 = 5$$

$$16 - 16x_2 + 4x_2^2 + x_2^2 = 5$$

$$5x_2^2 - 16x_2 + 11 = 0$$

$$(5x_2 - 11)(x_2 - 1) = 0$$

$$x_2 = 1 \text{ 或 } x_2 = 11/5。$$

当 $x_2 = 11/5$ 时, $x_1 = 4 - 2(11/5) = -2/5$, 不满足 $-x_1 \leq 0$ 。

当 $x_2 = 1$ 时, $x_1 = 4 - 2(1) = 2$ 。

点 $(2, 1)$ 满足所有 $g_i(x) \leq 0$ 条件。

将 $x_1 = 2, x_2 = 1, \mu_3 = 0, \mu_4 = 0$ 代入 $\nabla_x L = 0$ 的方程:

$$2(2 - 3) + 2\mu_1(2) + \mu_2 = 0 \implies -2 + 4\mu_1 + \mu_2 = 0 \implies 4\mu_1 + \mu_2 = 2$$

$$2(1-2) + 2\mu_1(1) + 2\mu_2 = 0 \implies -2 + 2\mu_1 + 2\mu_2 = 0 \implies \mu_1 + \mu_2 = 1$$

$$\text{解方程组 } \begin{cases} 4\mu_1 + \mu_2 = 2 \\ \mu_1 + \mu_2 = 1 \end{cases} \text{ 得 } \mu_1 = 1/3, \mu_2 = 2/3.$$

$$\mu_1 = 1/3 \geq 0, \mu_2 = 2/3 \geq 0, \mu_3 = 0, \mu_4 = 0.$$

所有KKT条件满足。

K-T点为 $(x_1, x_2) = (2, 1)$ 。

1. 某企业的生产函数 $Q = 3K^{\frac{1}{3}}L^{\frac{1}{2}}$ ，它表示在资本投入和劳动投入分别为 K 和 L 时，某种产品的产出量为 Q 。若产品价格为 $P = 4$ ，要素投入价格分别为 $P_K = 4, P_L = 3$ ，试求该企业得到最大利润时要素投入水平。

$$\Pi(K, L) = P \times Q - (P_K \times K + P_L \times L)$$

$$\Pi(K, L) = 4(3K^{\frac{1}{3}}L^{\frac{1}{2}}) - 4K - 3L$$

$$\Pi(K, L) = 12K^{\frac{1}{3}}L^{\frac{1}{2}} - 4K - 3L$$

$$\frac{\partial \Pi}{\partial K} = 12(\frac{1}{3}K^{-\frac{2}{3}})L^{\frac{1}{2}} - 4 = 4K^{-\frac{2}{3}}L^{\frac{1}{2}} - 4 = 0$$

$$\frac{\partial \Pi}{\partial L} = 12K^{\frac{1}{3}}(\frac{1}{2}L^{-\frac{1}{2}}) - 3 = 6K^{\frac{1}{3}}L^{-\frac{1}{2}} - 3 = 0$$

$$4K^{-\frac{2}{3}}L^{\frac{1}{2}} = 4 \implies K^{-\frac{2}{3}}L^{\frac{1}{2}} = 1$$

$$6K^{\frac{1}{3}}L^{-\frac{1}{2}} = 3 \implies 2K^{\frac{1}{3}}L^{-\frac{1}{2}} = 1$$

$$\text{由 } K^{-\frac{2}{3}}L^{\frac{1}{2}} = 1 \text{ 得 } L^{\frac{1}{2}} = K^{\frac{2}{3}} \implies L = K^{\frac{4}{3}}$$

$$\text{由 } 2K^{\frac{1}{3}}L^{-\frac{1}{2}} = 1 \text{ 得 } 2K^{\frac{1}{3}} = L^{\frac{1}{2}} \implies L = 4K^{\frac{2}{3}}$$

$$K^{\frac{4}{3}} = 4K^{\frac{2}{3}}$$

$$K^{\frac{2}{3}} = 4$$

$$K = (K^{\frac{2}{3}})^{\frac{3}{2}} = 4^{\frac{3}{2}} = (4^{\frac{1}{2}})^3 = 2^3 = 8$$

$$L = 4K^{\frac{2}{3}} = 4(8^{\frac{2}{3}}) = 4((8^{\frac{1}{3}})^2) = 4(2^2) = 16$$

最大利润时要素投入水平为 $K = 8, L = 16$ 。